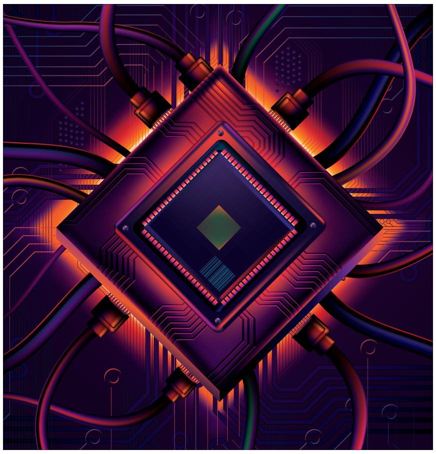
• Eta Devices

Eta devices have tried to reduce power consumption by developing a new design for amplifiers. Power amplifiers use power while sending digital data in two modes:standby and the output signal. This new design will lower the amount of power consumed in standby mode, while preventing the distortions in the digital signals that occur when moving from the low power standby mode to the high power output mode. While the design is still being lab-tested, the company aims to target the smartphone market soon. The main focus is its work on a smartphone chip, which will ultimately lead to a single power amplifier that can handle all of the different modes and frequencies used



Power efficient devices are the next step

by the various global standards. Once this tech is widely available, we can soon expect to find smartphones that consume half the power they're consuming today.

• Intel

Merrifield and Moorefield, two new Atom chips announced by Intel, promise faster performance and better battery life by powering down quickly after executing a task and are designed to work with many different operating systems.

• SAM L21 32-bit ARM family of microcontroller (MCUs)

Atmel, the San Jose-based company, introduced the new SAM L21 32-bit ARM family of microcontrollers, that consumes less than around 35 microamps of power per megahertz of processing speed while active, and less than 200 nanoamps of power when in deep sleep mode.

Though this still isn't sufficient to run an Ubuntu desktop, it has enough computing power and memory to run a real-time operating system with multiple programs, handle physical interfaces and stream media from a USB device or other external storage.

• Texas Instruments

Texas Instruments (TI) is developing chipsets for chargers that will increase the charging speed of your battery by almost 30%, which will in turn help increase the overall life of your battery. In short, you can expect a year-old battery to perform the same as a new one. TI's system should work with any standard lithium-ion battery in mobile phones today.

Eliminating batteries

What if you had access to technology that would eliminate the need for batteries? Ever thought of technologies that can power your devices 24x7?

Questions like these have fostered inventions as the following that will eliminate the need for storing power.

• Self-powered devices

Researchers from the Imperial College London are trying to create self-powered devices. By creating a material that can power devices, they expect to create the entire outer shell of the device with that material so that it can be powered by its body. Researchers are working to make it more durable, light-weight and energy efficient while still being cost effective.

• Wireless charging

With the increase in the number of wireless chargers, Ossia has added another one to the list, and it looks exciting. Ossia's 'COTA' technology can charge devices upto 30 feet of distance away via Wi-Fi or Bluetooth. The system needs a receiver and a charger. Once the receiver detects the charger, it sends signals to charge the device. It is not only inherently safe, but also highly efficient as the tracking beacons use only 1/10000th of the Wi-Fi's power and avoid anything that might absorb energy.

Natural batteries might be the answer

Natural charging

Yes, we have been using solar, wind and other natural sources of energy to power our houses for a while. But scientists are now working on using these to directly power our devices.

• Heat to electricity

MIT engineers are creating devices that can convert heat energy to electric energy. They can help charge our batteries with a temperature difference of 20 to 60 degree Celsius. Not just MIT, many others are also working towards the same goal and if it works out, batteries won't need to be plugged in for charging ever again.

• Smart clothing

We carry our phones in our pockets. What if our pockets could charge our phone? Researchers are trying to integrate carbon atoms into different types of yarns that will be used for making clothes. These smart clothing will have external powers and processing while only the sensors will be embedded in the clothes. Supercapacitors will be used to power the garments, and the sensors and touchscreen will power our phones.

• Muscle power

Nowadays, flexible nanogenerators are using our body muscles to power our batteries. A nanogenerator is a small, stamp-sized patch that attaches to your skin and converts the energy produced by muscle movements to electrical energy, which can then be used to power various devices.